

Regional Dissent: Do Local Economic Conditions Influence FOMC Votes?[†]

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US monetary policy decisions are made by the 12 voting members of the FOMC, seven of which inherently represent national-level interests. The remaining members, a rotating group of presidents from the 12 Federal Reserve districts, come instead from subnational jurisdictions. Does this structure have implications for the monetary policymaking process? We provide novel evidence that regional economic conditions influence the voting behavior of district presidents. Specifically, a regional unemployment rate that is 1 percentage point higher than the national level is associated with an approximately 9 percentage point higher probability of dissenting in favor of looser policy at the FOMC. (JEL E24, E32, E43, E52, E58)

The Federal Reserve System is composed of a Board of Governors in Washington, DC, and 12 Federal Reserve districts (see Figure 1). The system is responsible for upholding the dual mandate of price stability and full employment in the nation. To do so, the 19-member Federal Open Market Committee (FOMC) determines the best course of action for monetary policy. Seven members, including the chair, come from the Board of Governors. They are nominated by the US president and have voting rights at every FOMC meeting. The remaining 12 members are the presidents of the Federal Reserve districts. The FOMC employs a scheme where all district presidents participate in discussions but only five out of the 12 vote on monetary policy matters in any given year according to a rotating schedule. The New York Fed votes every year, Chicago and Cleveland vote every other year, and each of the other nine districts votes every three years.

In this paper, we investigate whether the regional component in the FOMC's voting structure has implications for monetary policy decisions. First, we construct a panel dataset of economic activity and FOMC voting across the 12 Fed districts. Second, we use our dataset to examine whether local economic conditions influence the voting behavior of district presidents. In brief, we run a district-level Taylor rule–like specification where the decision of district presidents to dissent at

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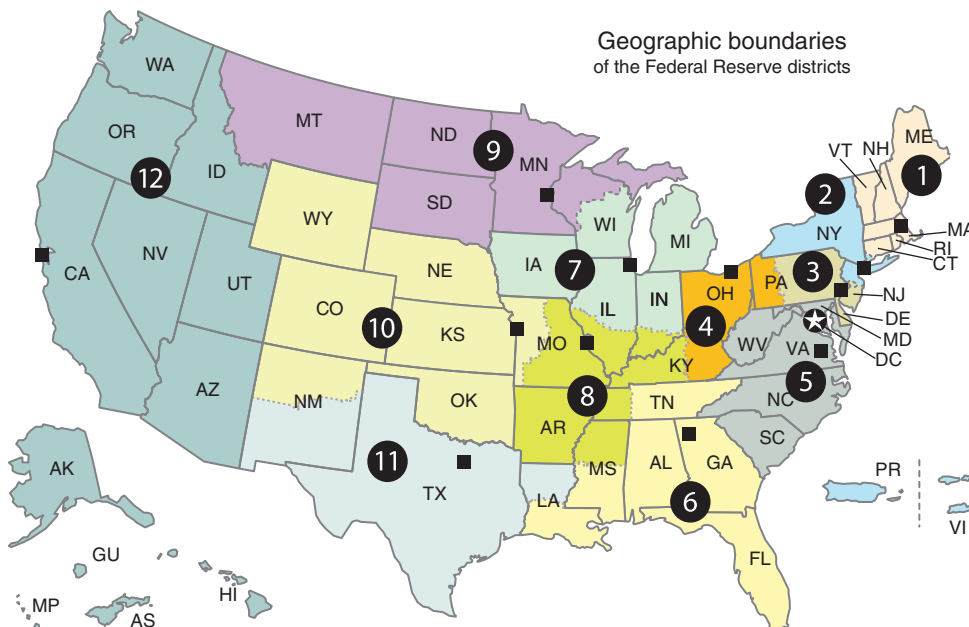


FIGURE 1. MAP OF THE FEDERAL RESERVE SYSTEM

Notes: Map of the geographic boundaries of the Federal Reserve System. The star depicts the Federal Reserve Board of Governors in Washington, DC. A square indicates the location of a given district's headquarters. The districts are: 1, Boston; 2, New York; 3, Philadelphia; 4, Cleveland; 5, Richmond; 6, Atlanta; 7, Chicago; 8, St. Louis; 9, Minneapolis; 10, Kansas City; 11, Dallas; and 12, San Francisco.

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the FOMC is regressed on district-level inflation and unemployment. Finally, we document how voting power scaled by economic activity is uneven across districts, which could result in policy biased toward overrepresented districts. In what follows, we describe these steps in further detail, along with our main results.

Section I begins with the history of the Federal Reserve System. It was designed to feature regional input; however, the explicit goals have always been national. Anecdotally, we find that district presidents will often refer to conditions in their own district, but whether their FOMC votes are biased toward their own district's conditions is ultimately an empirical question. Thus, we turn to constructing a district-level dataset. Districts are not necessarily divided along state lines. Consequently, we must construct macro variables from county-level data and aggregate up to the Fed district level. We obtain county-level labor force data from the BLS, nontradable inflation data from Hazell et al. (2022b), and FOMC voting data from Thornton and Wheelock (2014). We encode the presidents' dissents related to the stance of monetary policy as an indicator of agreement with the policy proposed by the Fed chair: 0 indicates agreement, -1 indicates a dissent in favor of tighter policy, and 1 indicates a dissent in favor of looser policy.

Our final dataset is a panel across Fed districts and FOMC meetings between 1990 and 2017. There are eight scheduled FOMC meetings a year, so we have a total of 224 scheduled meetings in our sample. Excluding New York, which never

dissents during our sample, there are four districts that vote each meeting. Therefore, we have 896 votes by non-New York district presidents during scheduled meetings in our sample. Out of these 896 votes, 72 are dissents in favor of tighter policy and 17 are for looser policy.

Section II analyzes whether local economic conditions influence FOMC voting. To study this issue, we run regressions in the spirit of a Taylor rule, where the dissent variable for a voting district president in a given FOMC meeting is regressed on regional unemployment and nontradable inflation as well as a rich set of fixed effects. The coefficient on nontradable inflation is insignificant.¹ More importantly, the coefficient on unemployment goes in the expected direction and is economically and statistically significant. A 1 percentage point higher district unemployment increases the probability of dissenting in favor of looser policy at the FOMC by roughly 9.2 percentage points. This is conclusive evidence that district presidents do not base their voting decisions solely on national evidence. However, we do not take a stance on the cause of this phenomenon. For example, presidents may have better information about their own district's economy and extrapolate this information to their beliefs about the nation's economy, or they may be partial toward their own district.

We perform a battery of robustness checks, including unscheduled FOMC meetings, using overall inflation instead of nontradable inflation, or lagging the inflation variable, without significant changes in the findings. Importantly, we also allow different responses of dissent to local conditions during and after the chairmanship of Alan Greenspan. During Greenspan's era, the coefficient on unemployment had approximately half the magnitude (5.6 instead of 9.2) and was nonsignificant. After his era, the coefficient increases (to 13.3) and becomes highly significant. This corresponds well with a notion in the literature that, under Greenspan, other FOMC members rarely used their votes to express dissent. Finally, we show that the results are similar when using ordered or penalized logistic regressions rather than ordinary least squares.

Section III discusses the implications of our findings. We show that districts have voting shares that are not proportional to their population or economic activity. Given the disparities in voting power and the fact that district presidents' votes reflect the economic conditions of their own district, monetary policy may be biased toward achieving the dual mandate in overrepresented districts. To get a sense of this effect, we perform a counterfactual exercise where all districts vote at each FOMC meeting and the weight of each vote is based on the labor force share of the corresponding district. Relative to the average predicted dissent across the districts actually voting at any given FOMC meeting, our labor force-weighted average predicted dissent is much smoother. Furthermore, there are certain periods where the "actual weighted" and "labor force-weighted" predicted average dissents differ substantially in sign and magnitude.

The role of local economic conditions in the FOMC's decision-making process has been the subject of a small but growing literature. Early work showed that local economic variables did not seem to impact the votes of district presidents. One of

¹The sign of the coefficient indicates that greater nontradable inflation in a district implies that its president is more likely to dissent in favor of tighter policy.

the first studies in this field, Tootell (1991), shows that regional economic performance had little to no effect on the voting patterns of district presidents from 1965 to 1985. Likewise, Jung and Latsos (2015) use forward-looking Taylor-type rules to estimate FOMC members' implicit policy reaction functions and find that few presidents exhibited regional biases during the Greenspan era.

Coibion and Goldstein (2012) find that dispersion in economic conditions across Fed districts matters for the setting of interest rates due to the nonlinearity of regional Phillips curves. However, they find no evidence that the relationship between interest rate decisions and regional heterogeneity is driven by FOMC members voting based on their regions of origin. Fos and Xu (2024) argue that, under some circumstances, district presidents might cast dissenting votes at the FOMC based on their own district's inflation.²

Some papers have studied disagreement at the FOMC using its members' spoken words rather than their observed votes. Meade (2005) uses FOMC transcripts rather than voting records to measure voiced dissent at FOMC meetings and finds that the rate of disagreement in internal discussions was quite high. Hayo and Neuenkirch (2013) use a probit model with regional and national macroeconomic variables to explain the content of Fed president speeches. The authors find that district presidents put relatively more weight on regional information when speaking within their home districts than when speaking outside of it. Bennani, Farvaque, and Stanek (2018) also find that regional conditions influence voiced policy preferences.

A small set of papers note that local economic conditions might indeed influence FOMC votes. Meade and Sheets (2005) find that economic conditions in the region of origin of FOMC participants (including board members) impact FOMC voting (without controlling for time fixed effects). Eichler, Löhner, and Noth (2018) use forward-looking Taylor-type rules with regional bank stability as an additional variable and find that regional bank stability had a significant effect on FOMC members' votes during Janet Yellen's term (2014–2018).

The literature has also proposed that FOMC chairs may amplify or dampen the effect of local economic conditions on policy preferences. Dominant leaders such as Greenspan may reduce the impact of local conditions by pressuring other members to follow their views. More open leaders such as Ben Bernanke, Yellen, or Jerome Powell may increase the impact of local economic conditions by allowing more diversity in opinions and dissent among members. This is consistent with our finding that the impact of economic conditions on dissent at the FOMC increases substantially after Greenspan.

Overall, the literature suggests that the influence of local economic conditions on FOMC decisions has historically been present in policy discussions, speeches, and general forms of voiced policy preference. We contribute to the literature by providing novel evidence that unemployment at the local level matters for the voting records of regionally affiliated FOMC participants.

²This refers to columns 4 and 8 of Table 5 in Fos and Xu (2024), including president and time fixed effects. Inflation is not significant in column 4, but it is significant at the 10 percent level in column 8, which uses the authors' "robust inflation measurement sample." Their result is conditional on analyzing "directionless" dissent, where one does not discern tighter dissents from looser ones.

I. Institutional Details and Data

A. History

The Federal Reserve Act of 1913 established the Federal Reserve System, consisting of the board and 12 regional banks. Given that the United States had recently emerged from the Banking Panic of 1907, the system was tasked with “furnish[ing] an elastic currency” and “establish[ing] a more effective supervision of banking.”³ The structure of the system emerged from a debate over the balance between centralization and decentralization. Some insisted on regional interests being represented in a decentralized manner. Representative Carter Glass, a cosponsor of the act, wrote: “In the United States, with its immense area, numerous natural divisions, still more numerous competing divisions, and abundant outlets to foreign countries, there is no argument, either of banking theory or of expediency, which dictates the creation of a single central banking institution, no matter how skillfully managed, how carefully controlled, or how patriotically conducted” (US Congress House Committee on Banking and Currency et al. 1913, 12). Others, such as Secretary of State Williams Jennings Bryan, believed that the federal government needed to be involved for coordination and oversight. A compromise was reached featuring both regional banks and a board consisting of individuals appointed by US presidents.

The FOMC was formed by the Banking Act of 1933, granting voting rights exclusively to Fed district presidents. The Banking Act of 1935 revised these protocols to include the Board of Governors. The final changes were instituted in 1942 to give the structure of 12 voting members with the current rotation scheme. The president of the New York Fed acts as the vice-chair of the FOMC and votes in every meeting. The historic economic significance of the Ohio River Valley and the manufacturing and transportation hub of Chicago resulted in the Cleveland and Chicago Feds’ participation in FOMC votes every other year. The remaining nine districts vote one year out of every three.

Following the end of the gold standard and the onset of stagflation in the early 1970s, Congress clarified the directives of the Fed by passing the Reform Act of 1977. This act states the current dual mandate: “The Board of Governors of the Federal Reserve System and the FOMC shall maintain long run growth of the monetary and credit aggregates commensurate with the economy’s long run potential to increase production, so as to promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates.”⁴

Throughout its history, the Federal Reserve System has had national goals. However, the regional banks were intended to bring local perspectives to policy discussions. Anecdotally, FOMC members often discuss district-level information and its value. In 1998, Governor Laurence H. Meyer said, “The presidents, in addition to having regional information, also tend to have real-time information about consumer spending, business investment, and wage and price developments, for example, gathered from speaking to firms in their districts” (Meyer 1998). As an example of local information influencing district presidents’ thinking, consider

³<https://www.federalreserve.gov/aboutthefed/officialtitle-preamble.htm>

⁴<https://fraser.stlouisfed.org/title/federal-reserve-reform-act-1977-1040>

Boston Fed President Eric Rosengren, who lamented that the number of cranes he could count on a short walk in Boston reached double digits in November 2015 and that “it is worth reflecting on the sustainability of that growth” (Rosengren 2015, 9). In April 2016, the Boston Fed’s Beige Book chapter noted that “construction activity increased significantly in greater Boston” and wages for skilled labor such as crane operators had risen substantially.⁵ Rosengren dissented in favor of tighter policy in September 2016. Similarly, Minneapolis Fed President Gary Stern was known to discuss car counts at the local malls (Rolnick, Runkle, and Fetting 1999). Additionally, anecdotal and qualitative assessments for each district are gathered in the Beige Book for scheduled FOMC meetings.

Overall, regional input was a feature of the system’s design; however, the system’s objectives are national. Anecdotal evidence suggests that district presidents emphasize their own district’s conditions in their assessments of the whole economy. Whether this can be detected in actual FOMC voting behavior requires a district-level dataset, which we now turn to constructing.

B. Data

Federal Reserve districts are not necessarily divided along state lines; some states (14 of them) are split between different districts. Consequently, we must construct macroeconomic variables from county-level data when possible and aggregate these observations to the Fed district level. We obtain county-level labor force data from the Local Area Unemployment Statistics program of the BLS (Bureau of Labor Statistics 2023). This dataset provides monthly labor force statistics (total number of people employed, unemployed, and in the labor force), not seasonally adjusted, for US counties starting in 1990. We aggregate these to the district level using a county-district correspondence from Tousey (2019a, b) and construct the unemployment rate by dividing the number of unemployed people in a district by the labor force in that same district.

We obtain inflation data from Hazell et al. (2022a, b), who construct state-level inflation series for the majority of US states between 1978 and 2017. They compute an overall category as well as tradable and nontradable subcategories. The data are quarterly, so we assign a given quarterly figure to all months in that quarter (which will typically cover two FOMC meetings).

We source county-level population data from IPUMS NHGIS (Manson et al. 2024) and Van Leuven (2022), both of which, in turn, ultimately come from the decennial US census. We get county-level GDP between 2001 and 2021 from the BEA Regional Economic Accounts (Bureau of Economic Analysis 2023). We obtain the history of district presidents from the Federal Reserve Bank of St. Louis (2024) and voting data from Thornton and Wheelock (2014, 2023). The voting dataset lists the Federal Reserve Board Governors and district presidents who dissent from the policy proposed by the Fed chair and categorizes dissents as being attributed to “tighter,” “looser,” or “other” justifications. We collect the district presidents’ dissents related to tighter and looser policy and encode them as an

⁵ www.federalreserve.gov/monetarypolicy/beigebook/beigebook201604.htm?boston

indicator of agreement with the proposed policy: -1 indicates a dissent in favor of tighter policy, 0 indicates agreement, and 1 indicates a dissent for looser policy.

Our final dataset (Bobrov, Kamdar, and Ulate 2024) is a panel across Fed districts and FOMC meetings between 1990 and 2017. There are eight scheduled FOMC meetings a year, so we have a total of 224 scheduled meetings in our sample. Excluding New York, which never dissents during our sample period, there are four remaining districts that vote each meeting. Multiplying the 224 meetings by four gives 896 votes by rotating Fed district presidents during scheduled meetings in our sample. Out of these 896 votes, there are 72 dissents for tighter policy and 17 dissents for looser policy.⁶ The remaining 807 votes therefore implicitly support the action proposed by the Fed chair.

II. Regional Dissent at the FOMC

A. Specification

In order to evaluate whether local economic conditions impact the votes of Fed district presidents at FOMC meetings, we run the following Taylor-type regression:

$$(1) \quad y_{it} = \delta_t + \zeta_{p(it)} + \alpha_0 \pi_{it} + \alpha_1 \pi_{it} \mathbf{1}\{t > 2006M1\} + \beta_0 u_{it} + \beta_1 u_{it} \mathbf{1}\{t > 2006M1\} + \epsilon_{it},$$

where i refers to the district ($1, 2, \dots, 12$); t refers to the FOMC meeting; y_{it} is the dissent variable for the president of district i at time t (-1 for a dissent in favor of tighter policy, 0 for agreement with the proposed policy, and 1 for a dissent in favor of looser policy); π_{it} and u_{it} are district i 's nontradable inflation and unemployment, respectively; δ_t is a set of time fixed effects; and $\zeta_{p(it)}$ is a set of president fixed effects.⁷

We always exclude the New York Fed given that it never dissented in our sample and its president is the vice-chair of the FOMC, an institutional position that makes dissent less likely. As Governor Meyer put it: "The president of the New York Fed simply won't dissent" (Meyer 2010, 29). However, the results are similar if we include New York. We use Driscoll–Kraay standard errors, which are robust to general forms of cross-sectional and temporal dependence with large time dimensions. We use four lags as would be suggested by Newey and West (1994). In some regression specifications, we include an indicator variable for FOMC meetings that occur

⁶Supplemental Appendix Table A.1 lists the dates of all dissents in favor of tighter policy along with the district dissenting, the district president's last name, the unemployment rate and inflation at the national and district levels, and an indicator that takes the value of 1 if the district unemployment was lower than the national level at the time of the dissent. Supplemental Appendix Table A.2 does the equivalent exercise but for dissents in favor of looser policy, and there the indicator takes the value of 1 if district unemployment was higher than the national level at the time of the dissent.

⁷This specification can be obtained from proposing a Taylor rule for a given district and a Taylor rule for the nation as a whole and then subtracting the two to obtain a specification in terms of "dissents" relative to the proposed policy on the left-hand side and deviations from national variables on the right-hand side. However, the national levels of unemployment and inflation would be absorbed by the time fixed effects, so they do not appear explicitly in equation (1).

TABLE 1—OLS REGRESSION EVIDENCE

	(1)	(2)	(3)	(4)	(5)	(6)
Constant	−0.990 (0.209)	−0.402 (0.240)	−0.369 (0.229)	−0.209 (0.298)	−0.392 (0.244)	−0.010 (0.166)
Unemployment	0.085 (0.030)	0.092 (0.039)	0.087 (0.038)		0.091 (0.039)	0.092 (0.040)
Nontradable inflation	−0.009 (0.026)	−0.001 (0.019)	−0.002 (0.020)			
Overall inflation					−0.003 (0.028)	
Lagged inflation						−0.000 (0.024)
Unemployment 1990–2005				0.056 (0.040)		
Nontradable inflation 1990–2005				0.002 (0.025)		
Unemployment 2006–2017				0.133 (0.065)		
Nontradable inflation 2006–2017				−0.010 (0.033)		
Observations	896	896	916	896	896	892
President FE	No	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Unscheduled	Drop	Drop	Keep	Drop	Drop	Drop
R^2	0.338	0.529	0.524	0.532	0.529	0.524
Adjusted R^2	0.116	0.335	0.328	0.337	0.335	0.327

Notes: This table presents the estimation results from regressions of the type in equation (1), where the dependent variable is an indicator of agreement with the policy proposed by the Fed chair: 1 indicates a dissent in favor of looser policy, 0 indicates agreement, and −1 indicates a dissent in favor of tighter policy. Columns 1–3, 5, and 6 do not include separate effects for the Greenspan and post-Greenspan periods, while column 4 does. The effects of inflation and unemployment after 2005 are $\alpha_0 + \alpha_1$ and $\beta_0 + \beta_1$, respectively. The effects of inflation and unemployment before 2006 are only α_0 and β_0 , respectively. Driscoll–Kraay standard errors are given in parentheses.

after January 2006 interacted with local unemployment and inflation; this allows us to separately identify the effects of local economic conditions during and after the term of Greenspan.

B. Baseline Results

Table 1 summarizes the results from regressions in the spirit of equation (1). In column 1, we present the simplest possible specification, where we only include time fixed effects, we do not include president fixed effects, and we do not separate the Greenspan and post-Greenspan periods. Additionally, we drop unscheduled meetings. The coefficient on nontradable inflation is not significant.⁸ More importantly, the coefficient on unemployment is economically and statistically significant and goes in the expected direction. A 1 percentage point higher district unemployment is associated with a probability of dissenting in favor of looser policy that is 8.5 percentage points higher.⁹

⁸The sign of the non–statistically significant coefficient indicates that greater nontradable inflation in a district implies that its president is more likely to dissent in favor of tighter policy.

⁹Supplemental Appendix Table A.3 reports the shares of dissents for tighter and for looser that “agree with our hypothesis,” in the sense that dissents for tighter are accompanied with a district unemployment that is lower than the nation’s and dissents for looser are accompanied with a district unemployment that is higher than the

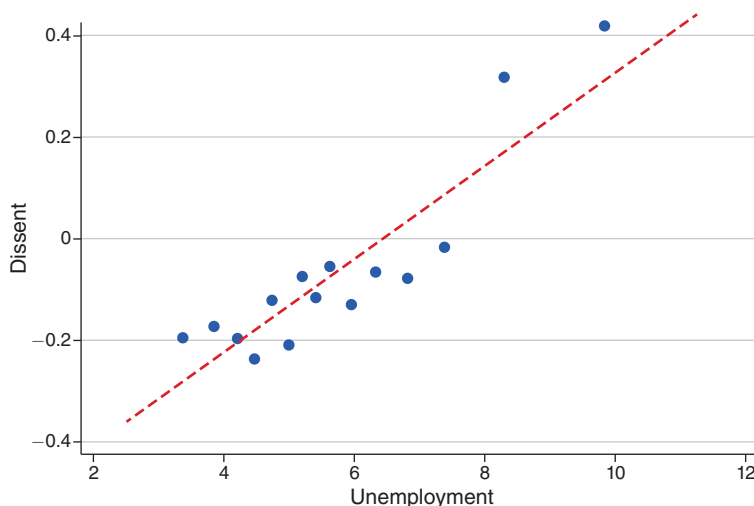


FIGURE 2. BINSCATTER OF REGIONAL UNEMPLOYMENT VERSUS DISSSENT

Notes: This figure plots a binscatter of regional unemployment against dissent using the specification in column 2 of Table 1 and the methods described in Cattaneo et al. (2019, 2024) (implemented using the Stata command “binsreg”). Importantly, the underlying information from which the binscatter is computed includes all of the 896 observations in our sample (not only the observations that have nonzero dissent). The specification in column 2 of Table 1, besides including regional unemployment, also includes time fixed effects, president fixed effects, and regional inflation.

In column 2 of Table 1, which is our baseline specification, we incorporate president fixed effects. This accounts for the fact that some presidents might be consistently more hawkish or dovish (Istrefi 2019; Bordo and Istrefi 2023; Hack, Istrefi, and Meier 2023). The estimated coefficient on nontradable inflation is near zero, but the coefficient on unemployment gets stronger and remains significant at the 5 percent level. Under this baseline specification, a 1 percentage point higher district unemployment is associated with a probability of dissenting in favor of looser policy that is 9.2 percentage points higher.¹⁰ Following the methodology of Cattaneo et al. (2024), Figure 2 displays a binscatter of regional unemployment versus dissent that helps to visualize the relationship between these variables in a nonparametric way. Including unscheduled FOMC meetings, as we do in column 3, does not alter the results in any significant way.

Note that part of the reason we find a significant response to regional unemployment but not to regional nontradeable inflation might be due to the fact that our measure of regional inflation is less precise than our measure of unemployment for several reasons. First, our inflation measure is at the state level rather than the county

nation’s. The table reports a version of this for the raw data, along with a version that performs a Frisch–Waugh–Lovell decomposition on column 2 of Table 1 and reports the shares of residualized dissents that agree with our hypothesis when compared with residualized unemployment (among observations that had nonzero dissent before residualizing).

¹⁰Supplemental Appendix Table A.4 presents results from our baseline regression with alternative measures of dissent. Using a simple indicator for dissent for looser, we find that higher district-level nontradable inflation decreases dissents for looser (p -value = 9.1 percent); however, this is based on a relatively small sample of 17 dissents in favor of looser policy.

level, so aggregation to the Fed district level is imperfect. Second, only 35 states are present in the Hazell et al. (2022a) data, and data from the available states is not necessarily representative within each of those states (as it comes only from certain big cities). Third, inflation is measured at the quarterly frequency rather than at the monthly one. For all of these reasons, there could be attenuation bias in the coefficient we estimate for regional inflation. We have used the best measure of inflation available to us and have also tried several others, but such alternative measures have their own drawbacks and do not significantly change any of our results.

C. Robustness

Within the FOMC voting literature, there is a notion that some FOMC chairs exhibited greater influence on their committees than others. Narrative accounts illuminate the power that Chairman Greenspan wielded to limit dissenting votes at the FOMC. To test this hypothesis, column 4 of Table 1 splits our sample into the Greenspan era and the post-Greenspan era.

In the Greenspan era, the unemployment coefficient is smaller than in our baseline specification (5.6 percentage points) and nonsignificant. By contrast, in the post-Greenspan era, the unemployment coefficient is higher than in our baseline specification, at 13.3 percentage points, and is highly significant. Overall, these findings support the notion that either Chairman Greenspan discouraged dissent or, for some other reason, district presidents did not take into consideration regional economic conditions when voting at the FOMC.¹¹

During the Greenspan period, dissenting votes by district presidents were rare. So it is not surprising that earlier literature did not reveal a significant relationship between voting behavior and local economic conditions. However, since 2006, there has been a rise in voting dissents. Specifically, our sample contains only 33 dissents between 1990 and 2005, for an average number of dissents per year of roughly 2 (0.25 dissents per FOMC meeting). The remaining 66 dissents in our sample happen between 2006 and 2017, for an average of 5.5 dissents per year (0.7 dissents per meeting).

Next, we conduct our baseline regression using alternative measures of inflation. Column 5 uses overall inflation rather than nontradable inflation, and column 6 uses lagged overall inflation. Both approaches yield estimates very similar to the baseline.

The qualitative results are largely not driven by any specific president. Supplemental Appendix Figure A.1 reports the coefficient on unemployment and its confidence interval for leaving out one district president at a time. All point estimates are positive. Excluding Evans, Lacker, Minehan, Moskow, or Yellen brings the unemployment coefficient above 10 percent. While most estimates are significant at the 5 percent level, there are two exceptions. Excluding Hoenig results in a *p*-value of 12.5 percent, and excluding Stern results in a *p*-value of 8.5 percent. These two presidents each account for 56 observations, the most of any president in our sample.

¹¹ To illustrate the potential sway Greenspan had over his committee, Supplemental Appendix Figure A.3 reproduces an excerpt from an FOMC transcript during his chairmanship.

TABLE 2—ORDERED AND PENALIZED LOGIT EVIDENCE

	Dissent (1)	Tighter (2)	Tighter (3)
<i>Panel A. Odds ratio</i>			
Unemployment gap	2.674 (0.760)	0.347 (0.119)	
Nontradable inflation gap	0.933 (0.186)	0.912 (0.186)	
Unemployment			0.318 (0.190)
Nontradable inflation			0.805 (0.259)
<i>Panel B. Unemployment margins</i>			
Tighter	−0.054 (0.016)		
Agreement	0.038 (0.013)		
Looser	0.016 (0.006)		
Unemployment gap		−0.073 (0.023)	
Unemployment			−0.112 (0.057)
Logit type	Ordered	Penalized	Penalized
Observations	896	896	896
President FE	Yes	Yes	Yes
Time FE	No	No	Yes
Unscheduled	Drop	Drop	Drop

Notes: This table presents estimated odds ratios (panel A) and average marginal effects (panel B) from ordered and penalized logit regressions. Column 1 is an ordered logit with our standard dissent variable as the dependent variable. Columns 2 and 3 present penalized logits (Stata command “firthlogit”) with the dependent variable being an indicator that is equal to 1 for dissents in favor of tighter policy and 0 otherwise. Gaps are defined as the deviation of the district-level economic variable from the respective national aggregate.

Our findings are robust to using logit specifications rather than the ordinary least squares baseline. Table 2, column 1 presents the estimates of an ordered logit. We are unable to include both time and president fixed effects, since there is “quasi-separation” in the data.¹² So, we use unemployment and nontradable inflation gaps from the aggregate to control for national conditions and include only president fixed effects. A 1 percentage point increase in the district unemployment gap significantly decreases the chance of dissenting in favor of tighter policy by 5.4 percentage points while significantly increasing the probability of agreeing with the chair by 3.8 percentage points and the probability of dissenting in favor of looser policy by 1.6 percentage points (all measured as average marginal effects).

This asymmetry between the three average marginal effects can be explained as follows. First, the support region of predicted values that corresponds to agreement with the policy proposed by the chair is large. Second, the majority of president fixed

¹² Many presidents never dissent, and at many meetings there are no dissents. Ordered logit iteratively drops observations until there is variation in each time and president, leaving limited observations.

effects put their respective presidents closer to the cutoff that separates agreement from dissents in favor of tighter policy than to the cutoff that separates agreement from dissents in favor of looser policy. Consequently, an increase in the unemployment gap is more likely to move presidents away from dissents in favor of tighter policy and into agreement with the chair than it is likely to move them away from agreements with the chair and into dissents in favor of looser policy.

Next, we use penalized logit regressions. This allows us to include time and president fixed effects despite quasi-separation.¹³ However, penalized logit is only for binary outcomes, so we alter our dependent variable to be an indicator for dissent in favor of tighter policy. Column 2 uses district-level gaps and only president fixed effects. In response to a 1 percentage point increase in the district unemployment gap, the probability of dissenting for tighter policy significantly decreases by 7.3 percentage points. Column 3 uses all regressors from our linear baseline: unemployment, nontradable inflation, and president and time fixed effects. In response to a 1 percentage point increase in district unemployment, the probability of dissenting for tighter policy significantly decreases by 11.2 percentage points. Overall, the ordered and penalized logit results concur with our baseline linear results.

One outstanding concern is that regional shocks may simultaneously affect several districts such that some voting and nonvoting districts may have systematically correlated conditions. In this case, our results could be driven not only by district conditions but by broader regional ones. We address this by using a placebo test presented in Supplemental Appendix Figure A.2. We create hypothetical samples where we pair each voting district i with a random district j 's economic data, run our baseline regression, and plot the density of the estimated unemployment coefficients. The distribution is roughly symmetric around zero, indicating that district presidents do not dissent more often in favor of looser policy if another randomly assigned district has high unemployment. We find our baseline unemployment coefficient of 0.092 to be larger than the corresponding coefficient in 98.5 percent of placebo samples. This suggests that not only are the results not driven by noise but that we are specifically capturing the conditions in the district of the voting president.

D. Discussion

While the regression in equation (1) captures correlations and not necessarily causal relationships, it is worth emphasizing that, even then, our results allow us to conclude that district presidents are not making decisions by considering national outcomes exclusively. This is the case because any such national influences on FOMC dissents would be absorbed by the time fixed effects, even if they were nonlinear and different than the traditional Taylor rule variables (i.e., inflation, unemployment, or the output gap).

¹³The penalized logit estimation procedure penalizes the log-likelihood by one-half of the logarithm of the determinant of the information matrix. The penalty term is minimized when the coefficient estimates are zero, effectively preventing coefficient estimates (e.g., our time or president fixed effects) from exploding. Therefore, it allows us to still include both time and president fixed effects, while traditional logit does not. See Firth (1993) for the initial discussion of this penalization as a method to reduce small sample bias from maximum likelihood estimates and Heinze and Schemper (2002) for extending the approach to the case of logit regressions with data that features quasi-separation. Lasso or ridge regression are other penalization techniques that serve a similar purpose but have an additional free parameter.

While we cannot necessarily pinpoint why district presidents act as if regional conditions matter to them when voting at the FOMC, we proceed to discuss some options. First, district presidents may inherently care more about their own district than about the nation as a whole, perhaps because they are accountable to their local board of directors, which is impacted primarily by local economic conditions. It is also possible that they are impacted by the revolving-door phenomena whereby they have worked for or want to work for firms in their district (Blanes i Vidal, Draca, and Fons-Rosen 2012; Lucca, Seru, and Trebbi 2014). Second, district presidents may have better or more up-to-date knowledge about their own district than about the whole nation due to information frictions (Mankiw and Reis 2002; Sims 2003), and they may believe that their local information is representative of the national economy. Third, district presidents may believe that the sectoral composition of their district is representative of the national economy and may receive more signals from sectors that are important in their district (e.g., oil for the Dallas Fed, finance for the NY Fed, etc.). Regardless of the cause, we can assert with a 95 percent confidence level that district presidents consider regional unemployment when voting at the FOMC.

III. Implications

Due to the structure of FOMC voting, Fed districts have heterogeneous voting shares. The allocation of voting power across districts does not necessarily correspond to the population or economic significance of the regions they represent, which may create a democratic deficit in monetary policy decision-making.

The top panel of Figure 3 depicts recent GDP, labor force, and population shares of each Fed district. The district voting shares differ significantly from the economic and population shares of each district. For instance, the Cleveland Fed district has twice as much voting power as its GDP share would imply (10 percent voting share compared to a 5 percent GDP share). By contrast, the San Francisco Fed district has about a quarter of the voting power relative to what its GDP share would suggest (6.66 percent voting share compared to a 24 percent GDP share).

Given that district presidents' votes are influenced by regional conditions, monetary policy may be biased toward achieving full employment and stabilizing inflation in the districts that are overrepresented. To get a glimpse of the potential implications, we now turn to comparing how FOMC voting might have behaved if all Fed districts (instead of just 5 out of 12) voted at the FOMC.

In order to predict how district presidents would have voted based on our econometric specification, we start with our baseline model in column 2 of Table 1. However, we exclude the time fixed effects, as we wish to construct an aggregate counterfactual that would be confounded by them. Without time fixed effects, we use deviations in the regional variables from their national aggregates. We arrive at the following regression:

$$(2) \quad y_{it} = \delta + \zeta_{p(it)} + \alpha_0(\pi_{it} - \pi_t) + \beta_0(u_{it} - u_t) + \epsilon_{it}.$$

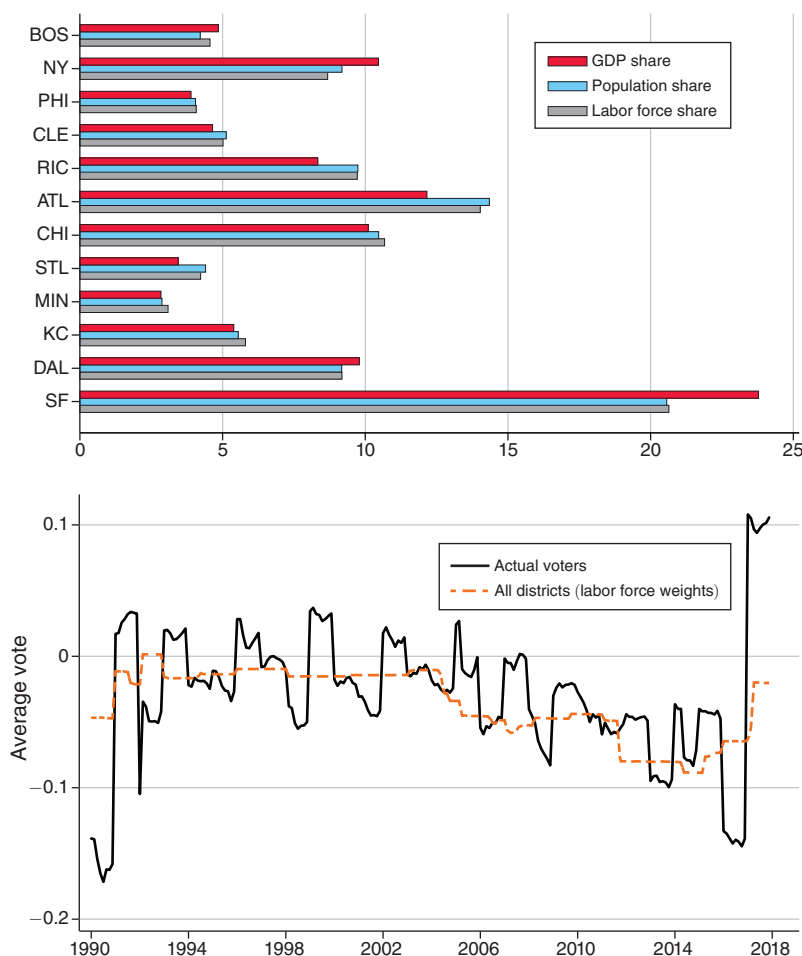


FIGURE 3. REPRESENTATION AND AVERAGE PREDICTED VOTES

Notes: The top panel displays the shares of GDP, population, and labor force that each of the Fed districts accounted for in 2020 (population) or 2021 (GDP and labor force). The bottom panel plots the predicted value of dissents computed as described around equation (2), either averaged across only the districts that voted in each FOMC meeting (solid black line) or across all 12 Fed districts using labor force weights (dashed orange line).

We estimate equation (2) for the sample in column 2 of Table 1 and use these estimated coefficients to obtain the predicted value of dissent for all districts. Specifically, the predicted dissent of district i at time t is $\hat{y}_{it} = \hat{\delta} + \hat{\zeta}_{p(it)} + \hat{\alpha}_0(\pi_{it} - \pi_t) + \hat{\beta}_0(u_{it} - u_t)$.

Having obtained the predicted dissent value for each district at each FOMC meeting, the bottom panel of Figure 3 aggregates these values across districts in two different ways. The black line aggregates the values of the five districts that actually voted at the corresponding FOMC meeting, each with a weight of one-fifth. By contrast, the dashed orange line aggregates the values of predicted dissent across all districts using time-varying labor force weights. As throughout the paper, dissents for tighter take the value of -1 , agreements take the value of 0 , and dissents for looser take the value of 1 . Therefore, the black line being at 0.1 , as it is for most of

2017, indicates that at each meeting that year there was around a 50 percent chance that one voting district would have dissented in favor of looser policy.

The crucial feature of the bottom panel of Figure 3 is the substantial fluctuation that the black line displays at the yearly frequency, as the FOMC voting rotation shifts between districts that are at different business cycle stages compared to the US average (and have presidents with different hawkish/dovish tilts, as described in Hack, Istrefi, and Meier 2023). By contrast, the orange line is much more stable and does not fluctuate gratuitously due to arbitrary annual voting rotation changes.

Consider, as an illustrative example, the case of 1999. The predicted average vote with the actual voting shares (solid black line) was around 3.5 percent, indicating an average preference for looser policy coming from the regional Feds at the FOMC. This positive value was due mostly to the fact that the Dallas Fed and the New York Fed were in the voting rotation that year and had higher unemployment than the US average (and, in the case of President McTeer of Dallas, a relatively dovish president), while Kansas City and Richmond were excluded from the voting rotation and had much lower unemployment than the US average (and, in Hoenig and Broadus, relatively hawkish presidents). The predicted average vote across all districts with labor force weights (dashed orange line) was instead around -1.5 percent, indicating that if all districts had been considered, there would have instead been a preference for tighter policy stemming from the regional portion of the FOMC.

While we do find qualitative differences between the actual weighted and labor force-weighted predicted votes, this exercise likely underestimates the magnitude of regional influence. Regional dissent could appear in voice rather than votes. Additionally, the policy proposed by the chair (or the language in the FOMC statement) could have been altered before voting to prevent an official dissent by a voting district president whose district's economic conditions were particularly different from the national ones.

IV. Conclusion

We contribute to the FOMC decision-making literature by providing evidence that regional economic conditions influence the voting behavior of regionally affiliated FOMC members significantly. Specifically, we find that a 1 percentage point higher district unemployment rate increases the likelihood that the president associated with that district dissents in favor of looser policy at the FOMC by more than 9 percentage points. Overall, this suggests that the district and voting structure of the FOMC results in monetary policy that may be biased toward districts whose share of FOMC votes exceeds their share in the US population (or economy). While we do not take a stand on why presidents' votes reflect their own districts' economic conditions, this would be an interesting avenue for future work.

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